

OBJECTIVE

Mechatronics engineering student with proven experience in AI/ML deployment, data pipeline engineering, and intelligent automation, aiming to join maids.cc as a Business Analyst to design scalable workflows, build AI-powered chatbots, and deliver data-driven operational tools across a 26,000+ employee ecosystem.

TECHNICAL SKILLS

AI & Automation: Python, PyTorch, OpenCV, LangChain-compatible REST APIs, Flask, Streamlit

Data & Analytics: SQL, Azure Databricks, Apache Spark, MLflow, PostgreSQL, Feature Engineering

Tools & Platforms: Git, Linux, MATLAB, JIRA (familiar), Notion (familiar), Miro (familiar)

EDUCATION

Egypt Japan University of Science and Technology (EJUST)

Burj El-Arab, Alexandria

B.Eng. in Mechatronics Engineering — GPA: 3.54/4.0

Sep. 2023 – Expected May 2027

- Relevant Coursework: Data Analysis, Algorithms, Artificial Intelligence, Operating Systems, Electronic Circuits

EXPERIENCE

Samsung Innovation Campus (SIC)

Remote

AI/ML Intern

Sep. 2024 – Nov. 2024

- Architected end-to-end ML pipelines using AlexNet and custom CNNs, then deployed real-time inference models via Flask and Streamlit with RESTful APIs — directly paralleling the chatbot and automation tooling at maids.cc.
- Drove measurable accuracy gains through iterative feature engineering and data preprocessing on complex fashion datasets, producing reproducible, well-documented pipelines.

Digital Egypt Pioneer Initiative (DEPI)

Hybrid

Data Engineer Trainee

Apr. 2024 – Oct. 2024

- Built scalable data pipelines on Azure Databricks and Apache Spark for datasets exceeding 100 GB, optimizing complex SQL queries to deliver analytics-ready outputs for downstream decision-making.
- Integrated MLflow MLOps workflows for experiment tracking and model versioning, enabling auditable, reproducible analytical processes across cloud-based architectures.

Tech Maker (ITIDA EME)

Onsite

Mechatronics & Embedded Systems Intern

Sep. 2024 – Nov. 2024

- Developed a real-time Battery Management System (BMS) with live dashboarding via MQTT and HTTP, demonstrating IoT-to-operations data integration for continuous system monitoring.

PROJECTS

AgriBot — Autonomous Agricultural Robot | *ROS2, Python, OpenCV, PostgreSQL, AI/ML*

Mar. 2025

- Designed a relational database backend for dynamic task assignment and field-zone tracking, automating operational workflows across software, sensor, and data layers — mirroring enterprise KPI and automation systems.
- Built a digital twin GUI for real-time field monitoring and robot state visualization, enabling remote operational oversight comparable to a live operations dashboard.

3-Wheeled Mecanum Mobile Robot | *micro-ROS, OpenCV, ESP32, Digital Twin, Simulation*

Jan. 2026

- Integrated a 7-parameter soil sensor array and an OpenCV plant-detection module to automate condition assessment and data-driven task routing, demonstrating sensor fusion and intelligent process automation end-to-end.

Autonomous Guided Vehicle (AGV) System | *Python, Raspberry Pi, OpenCV*

Jan. 2024

- Built a vision-guided AGV with QR-code localization and a 2-DOF robotic arm for autonomous pick-and-place, delivering a full perception-to-action automation pipeline without third-party integration.

CERTIFICATIONS

Certified SolidWorks Professional (CSWP), Dassault Systèmes — Dec. 2025

Certified SolidWorks Associate (CSWA), Dassault Systèmes — Sep. 2025

LEADERSHIP & VOLUNTEER

Digital Egypt Marvels Initiative (DEMI)

Online

Programming Instructor

Jun. 2024 – Sep. 2024

- Designed and delivered structured programming lessons to 20+ students aged 10–12, translating complex technical concepts into clear, actionable curricula.

IEEE Robotics and Automation Society (RAS) — EJUST Chapter

EJUST

Electronics & Robotics Software Volunteer

Oct. 2024 – Jun. 2026

- Mentored junior members on embedded electronics and hardware integration; maintained 3D printing lab equipment for continuous prototyping cycles.